

Composing an Authority Layer with a Sandbox Runtime: BULL on NVIDIA OpenShell

An empirical qualification against a live OpenShell v0.1.2 gateway

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Abstract

NVIDIA OpenShell runs agents in sandboxes. Each sandbox enforces a declarative network and filesystem policy and emits OCSF telemetry. It also exposes two extension points: a gateway interceptor on control-plane RPCs and a supervisor middleware on each proxied HTTP request. BULL is an authority layer. It decides from signed grants, content provenance and single-use human approvals whether an agent's action is permitted, and it records every decision in a hash-chained ledger. We built BULL as an OpenShell interceptor and middleware and tested one invariant against a real gateway built from source:

EFFECT = BULL_authorized $\wedge$ OpenShell_policy_allowed, and a missing or failing BULL means no effect.

We used a 10-case proof suite plus two extra cases (C3b, C5b). Each case ran in two phases, OpenShell alone (A) and OpenShell with BULL registered fail-closed (B). We judged effects at the destination: a request counts as "blocked" only if the upstream server never received it. **All 26 checks passed in the final run.** Composition cost a median **+8.1 ms** per GET and **+11.0 ms** per POST (n = 200 each), with no significant change to sandbox creation time.

The experiment also produced findings that matter to anyone deploying either system:

1. OpenShell's L7 rules default to enforcement: audit, which logs violations and then **allows** them. We measured this directly in C3b-A.
2. OpenShell v0.1.2 hands middleware the scheme https for plaintext HTTP relays and never populates `originating_process`.
3. BULL's ledger append re-verifies the whole chain, which makes latency grow with ledger size.

We fixed six integration defects in BULL during the run; all are listed in §6.

1. Motivation

A sandbox answers *where* code may reach. An authority layer answers *whether this particular action, now, given where its inputs came from, is authorized and by whom*. Each is weak alone:

- **OpenShell without an authority layer.** It has no notion of content provenance. A sandbox that reads attacker-controlled text can POST to any endpoint its policy allows (C4-A and C7-A below). Its policy can be widened by anyone who can call `UpdateConfig` (C5-A).
- **BULL without a sandbox.** BULL's decisions bind only on traffic that passes through it. A native tool that skips BULL's proxy is outside BULL's view.

The question is whether both can be composed so each covers the other's gap, without the composition opening a new bypass. The failure mode we most wanted to exclude is fail-open:

BULL down, and effects happening anyway.

2. System under test

2.1 OpenShell extension points used

Extension	Where it runs	What BULL does there
Supervisor middleware (SupervisorMiddleware/EvaluateHttpRequest, phase PRE_CREDENTIALS)	Called by each sandbox's supervisor for each proxied HTTP request, after OpenShell's own L7 policy and before credential injection	Returns ALLOW or DENY. It can only narrow what OpenShell's policy already allowed
Gateway interceptor (GatewayInterceptor/Evaluate)	Called by the gateway on control-plane RPCs	modify_operation: attaches BULL middleware (onError: fail_closed) to every host in a new or replacement policy. validate: refuses unmediated sandboxes, audit-mode L7 rules, hosts outside the signed ceiling, and any unapproved widening

Governed RPCs:

- CreateSandbox and UpdateConfig: modify_operation and validate.
- AttachSandboxProvider, ApproveDraftChunk, ApproveAllDraftChunks and EditDraftChunk: validate only.

Both extensions are registered with failure_policy = "fail_closed".

2.2 BULL components added (src/bulldog/openshell/)

Module	Role
grant.s.py	Per-sandbox grants (bull-openshell-grants-v1) inside the HMAC-signed BULL policy bundle: host ceiling, capabilities, trusted host:port sources
authority.py	Decisions. HTTP method maps to capability; provenance becomes INTERNET once a sandbox has read an untrusted source; ESCALATE yields a pending approval bound to the exact method, URL and body hash, consumed once. Control-plane checks cover widening, the ceiling, audit mode and credential attach. State persists across restarts
services.py	gRPC servicers for both extension protocols (PeerMetadata 1.0, contract capabilities)
server.py	Process entry point, plus a private operator socket (pending, approve)
ocsf.py	Parses OpenShell OCSF shorthand as served by openshell logs, and joins it with the BULL ledger and observed effects
backend.py	Health and extension-presence probe (require_strict)

3. Method

3.1 Environment

Item	Value
OpenShell	v0.1.2 (commit 6648bd0), built from source with Rust 1.95.0. Gateway, CLI and supervisor are glibc builds; the sandbox runtime is a static musl build
Compute driver	Docker 29.3.1 (image_pull_policy = "never")
Host	Linux 6.18.44, 4 vCPU, 15 GiB, cloud container

Item	Value
Sandbox base image	ubuntu:24.04 plus curl and python3, run as uid 1500
BULL	Python 3.11, grpcio 1.84, protobuf 7.36
Transport	Middleware over plaintext gRPC on the Docker bridge (allow_insecure_transport = true); interceptor on a Unix socket

Deviations from a stock OpenShell install. The sandbox's network policy blocks `ghcr.io`, `nvcr.io` and GitHub release downloads, so we could not use the official installer or published images. We built all images locally from the upstream Dockerfiles, with two changes:

- **Supervisor base image.** The supervisor image uses `gcr.io/distroless/cc-debian13` instead of the pinned `distroless/base-nossl-debian13`, because a locally built supervisor needs `libgcc_s`.
- **Host alias.** The host maps `host.openshell.internal` to the Docker bridge (172.17.0.1) in `/etc/hosts`, so the gateway, which runs on the host, can reach the middleware at the same address the supervisors use.

Policies declare `allowed_ips`: `["172.17.0.1/32"]`, because OpenShell's SSRF guard otherwise blocks the private bridge address.

3.2 Topology

Three host-side HTTP services act as destinations and record every request that reaches them:

- **docs:** trusted by BULL.
- **news:** untrusted; it returns a prompt-injection string.
- **api:** the effect target.

OpenShell's policy allows:

- GET `/**` on docs and news
- POST `/v1/**` on api

All three endpoints set enforcement: enforce and limit the binary to `/usr/bin/curl`.

BULL grants:

Sandbox	Capabilities	Trusted source
coder	outbound + post	docs
reader	outbound only	docs
bench	outbound + post	docs

The host ceiling is `host.openshell.internal`.

3.3 Phases and judging

- **Phase A:** OpenShell only.
- **Phase B:** the same policies and requests, with both BULL extensions registered.

A case passes on effect evidence, meaning upstream receipts, never on client output alone. Phase A rows for BULL-specific cases record what OpenShell alone does. A phase A "pass" means the observation was made as expected, not that the property holds.

3.4 Latency

Inside one bench sandbox, a shell loop runs `curl -w %{time_total} 200` times for a trusted GET and 200 times for an allowed POST. This keeps sandbox exec overhead out of the samples. Sandbox creation time is the `openshell sandbox create --detach` wall time over 6 creations; it measures admission, including interceptor round-trips, not boot to Ready.

The harness is `tools/openshell_experiment.py`. It is fully automated and builds, runs, collects and tabulates.

4. Results

4.1 Proof suite (final run)

C as e	Property	A: OpenShell only	B: OpenShell + BULL
C 1	BULL ALLOW $\wedge$ OpenShell ALLOW $\rightarrow$ effect	200, reached	200, reached
C 2	BULL DENY $\wedge$ OpenShell ALLOW $\rightarrow$ no effect (reader has no post grant)	200, reached (no authority layer)	403, not reached (bull_denied)
C 3	BULL ALLOW $\wedge$ OpenShell DENY (POST /admin/delete, outside L7 rules) $\rightarrow$ no effect	403, not reached	403, not reached
C 3 b	Same L7 rules left at OpenShell's default enforcement	sandbox created; POST /admin/delete reached	creation refused (bull_l7_audit_mode)
C 4	ESCALATE: no effect before approval; exactly one after; none on reuse	200, reached (no approval concept)	403 $\rightarrow$ approve $\rightarrow$ 200 $\rightarrow$ 403 ; upstream count exactly 1
C 5	Unauthorized policy widening (new port on an allowed host)	applied (policy v3 loaded)	refused : approval ... required to widen policy with ['host.openshell.internal:9']
C 5 b	Sandbox policy naming a host outside the signed ceiling	(n/a)	refused (bull_outside_ceiling)
C 6	Provider credential never readable by the agent	canary not in sandbox env	attach refused (no credential.read grant); canary not in env
C 7	Internet-derived content, then POST, hits provenance control	POST after reading news reached	403, not reached (bull_approval_required, provenance internet)
C 8	Native bypass: <code>curl --noproxy</code> , a Python HTTP client, a raw socket to 1.1.1.1:443, UDP DNS to 8.8.8.8	--noproxy POST reached, but <i>mediated</i> (OCSF L7 event present); Python HTTP and raw socket EACCES; UDP blocked	--noproxy POST intercepted and escalated by BULL (not reached; OCSF shows both l7 and middleware); others blocked as in A
C 9	BULL process killed: egress and sandbox creation fail closed	(n/a)	POST 403 (middleware_failed: external_service_error), not reached; CreateSandbox refused ; after restart, trusted GET 200 with taint and approvals preserved
C 1 0	OpenShell events, BULL decisions and effects form one audit trail	8 L7 events, no authority record	9/9 BULL decisions matched to OpenShell middleware events and consistent; 0 effects without a BULL allow; the 1 orphan OpenShell event is the C9 fail-closed denial; ledger hash chain verifies

Totals: phase A, 12/12 observations as expected. Phase B, 14/14 properties held. Both include the two setup rows.

4.2 Audit correlation detail (C10, phase B)

Sandbox	Request	BULL	OpenShell L7	OpenShell middleware	Effect
coder	GET docs/guide	ALLOW	ALLOWED	ALLOWED	yes
coder	POST api/v1/records	ALLOW	ALLOWED	ALLOWED	yes
reader	POST api/v1/records	DENY	ALLOWED	DENIED (bull_denied)	no
coder	GET news/article (taints)	ALLOW	ALLOWED	ALLOWED	yes
coder	POST api/v1/records	ESCALATE → pending	ALLOWED	DENIED (bull_approval_required)	no
coder	same POST after approval	ESCALATE → approved (consumed)	ALLOWED	ALLOWED	yes (exactly once)
coder	same POST again	ESCALATE → pending	ALLOWED	DENIED	no
coder	POST api/v1/x (--noproxy)	ESCALATE	ALLOWED	DENIED	no
coder	GET docs/after-restart	ALLOW	ALLOWED	ALLOWED	yes

The L7 column is ALLOWED on every row. That is the point of the composition: OpenShell's policy alone would have let every one of these requests through.

OpenShell carries BULL's reason code into its own denial telemetry, for example `middleware_denied:bull-governance:bull_approval_required`. A SIEM that ingests only OCSF can therefore still see why BULL refused.

4.3 Latency

Per request, measured inside the sandbox; n = 200 per cell; values in ms.

Metric	Phase	Mean	Median	p95	Min	Max	σ
GET (trusted)	A	10.23	8.99	16.92	6.36	31.20	3.49
GET (trusted)	B	17.88	17.07	26.03	11.43	43.46	4.34
POST (allowed)	A	11.46	9.85	21.17	6.65	33.86	4.60
POST (allowed)	B	22.65	20.83	34.17	14.42	65.33	6.60

Sandbox creation, n = 6, in seconds:

Phase	Mean	Median	p95	Min	Max
A	0.914	0.925	0.939	0.811	1.026
B	0.986	0.995	1.010	0.908	1.058

Derived observations:

- **Per-request overhead.** Median +8.1 ms per GET and +11.0 ms per POST. From OCSF timestamps, OpenShell's L7 event precedes the middleware verdict by a median of 5.0 ms (range 1–7 ms, n = 10). That is the BULL round-trip as the supervisor sees it: plaintext gRPC over the bridge, the Python decision, and a synchronous ledger append.

- **Admission overhead.** About +70 ms mean per sandbox creation (two interceptor round-trips on a Unix socket). With $n = 6$ and $\sigma \approx 50\text{--}70$ ms, this is at the edge of the noise.
- **Drift with ledger size.** Phase B latency rises across the run: median of the first 50 versus the last 50 samples goes $14.9 \rightarrow 17.7$ ms (GET) and $19.5 \rightarrow 25.1$ ms (POST). Phase A stays flat ($9.1 \rightarrow 9.7$ and $10.3 \rightarrow 9.2$). BULL's AuditLedger re-verifies the entire hash chain before every append, so append cost is $O(n)$ in ledger length. This is the most likely cause; we did not isolate it. It also explains why POST overhead exceeds GET overhead: the POST loop ran second, against a longer ledger. A persistent tail-hash check would make appends $O(1)$.

Figures

All figures are generated from the evidence files by tools/build_openshell_paper_pdf.py.

Figure 1. Outcome matrix. Red in A = an effect OpenShell alone let through that the composed system blocks

A	observed	effect happened	observed	effect happened	effect happened	effect happened	n/a	observed	effect happened	observed	n/a	observed
B	held	held	held	held	held	held	held	held	held	held	held	held
	C1	C2	C3	C3b	C4	C5	C5b	C6	C7	C8	C9	C10

Figure 2. Per-request latency distributions (n = 200 per cell)

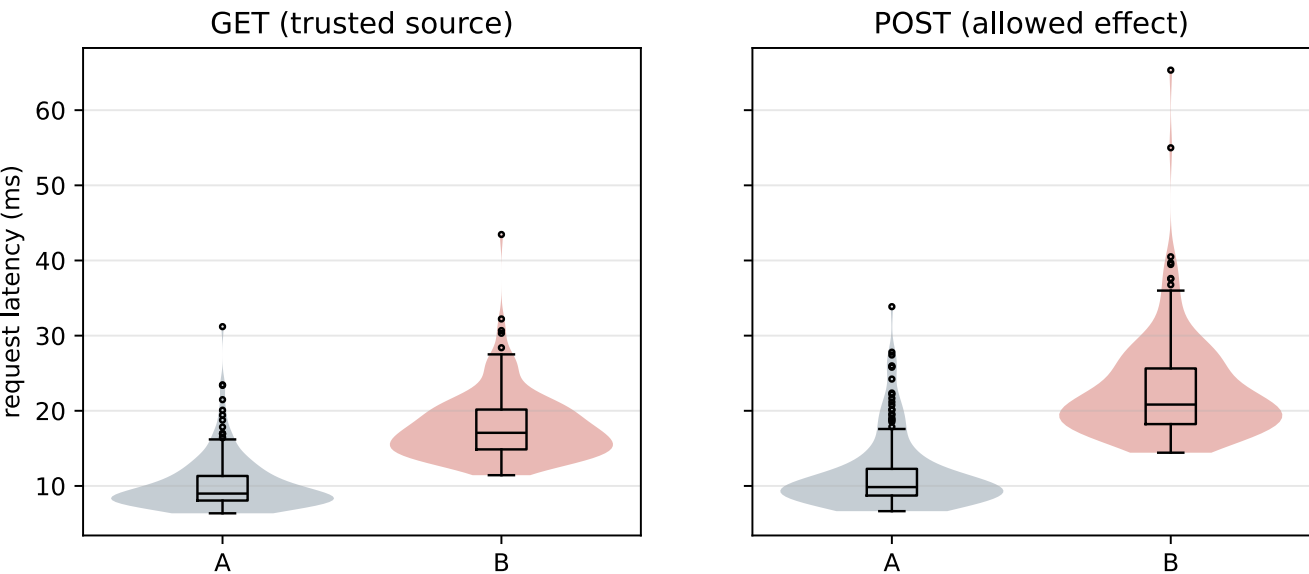


Figure 3. Empirical CDF of request latency

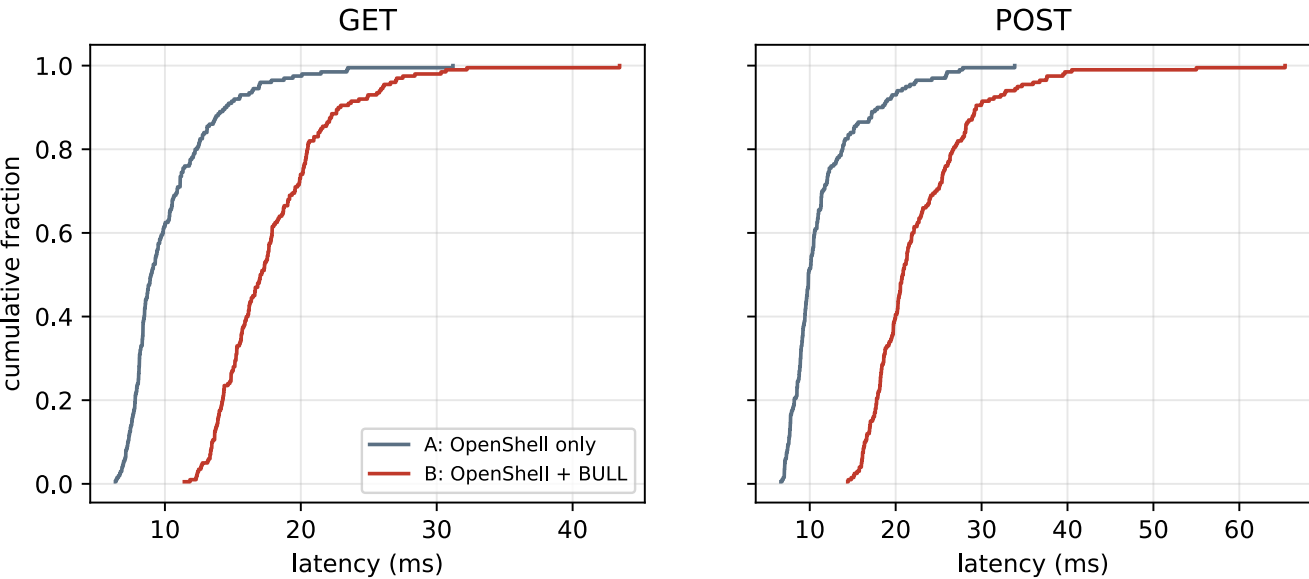


Figure 4. Latency over the run: phase B drifts upward as the ledger grows

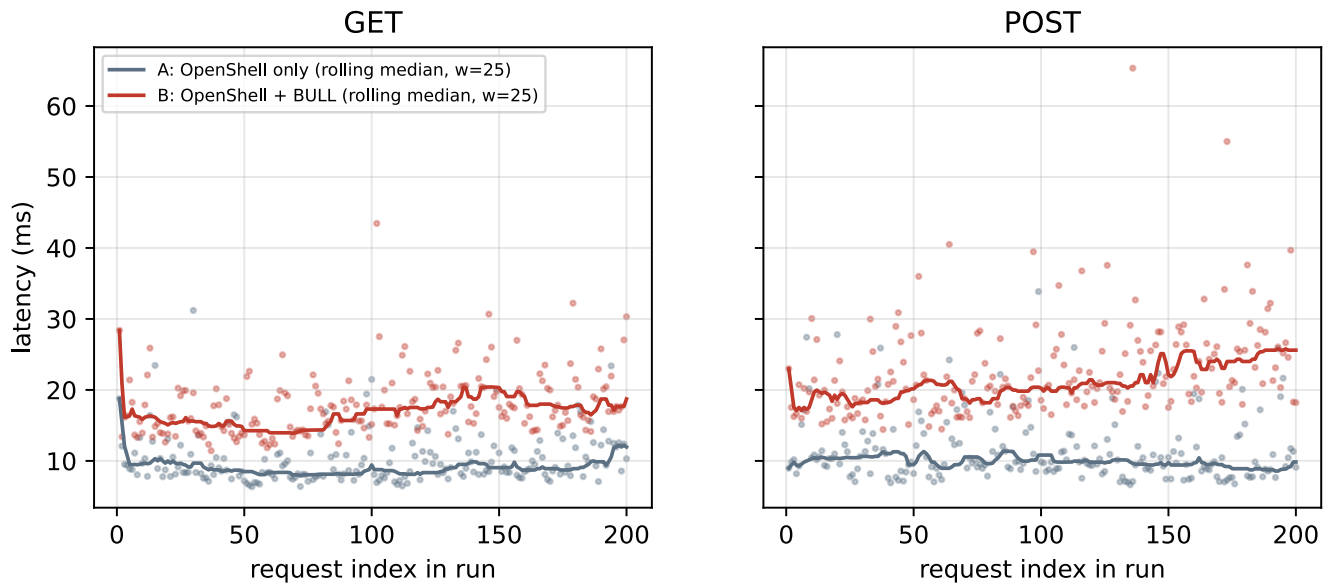


Figure 5. Sandbox admission time (n = 6; bar = median)

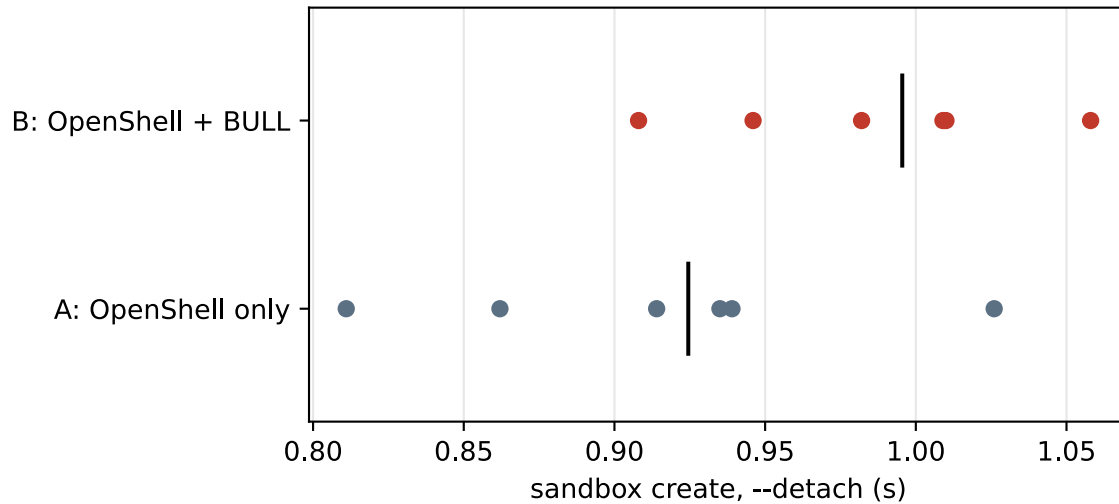
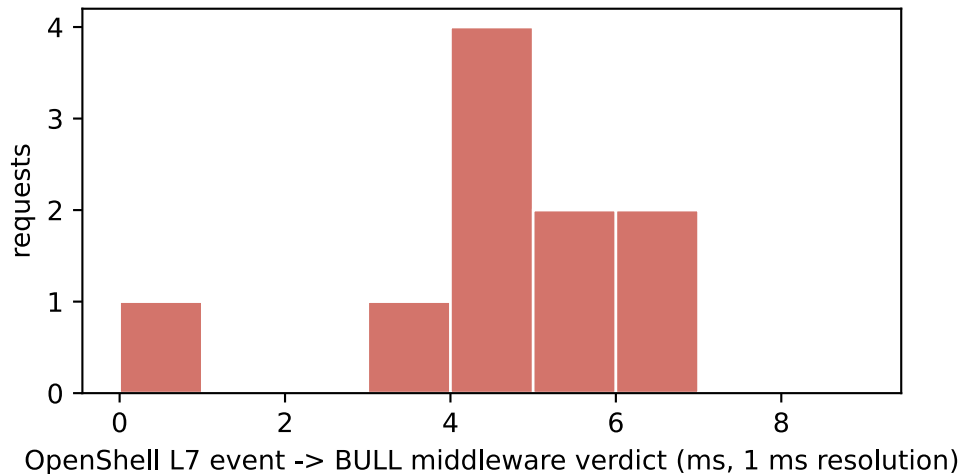


Figure 6. BULL round-trip seen by the supervisor (n = 10)



Statistical analysis (computed from latency-raw.json)

Table S1 gives the overhead as a median difference (B – A). The 95% confidence interval comes from a percentile bootstrap with 10,000 resamples and a fixed seed. Significance comes from a two-sided Mann–Whitney U test with normal approximation; the effect size is the rank-biserial $|r|$.

Metric	n per phase	Median A	Median B	Δ median (ms)	95% CI	U	z	p	r
GET	200	8.98	17.07	+8.08	[+7.12, +8.86]	2525	-15.11	< 1e-15	0.874
POST	200	9.85	20.83	+10.98	[+10.16, +11.72]	2470	-15.16	< 1e-15	0.877

Table S2 compares the median of the first 50 samples with the median of the last 50 in each run.

Metric	Phase	First 50	Last 50	Change (ms)
GET	A	9.07	9.68	+0.61
GET	B	14.89	17.67	+2.77
POST	A	10.33	9.20	-1.13
POST	B	19.49	25.13	+5.64

Table S3 gives the full descriptive statistics, with percentiles by linear interpolation. Latencies are in ms; creation times are in s.

Phase	Metric	n	Mean	Median	p5	p25	p75	p95	p99	Min	Max	σ
A	get_trusted_ms	200	10.228	8.985	6.965	8.058	11.329	16.917	23.382	6.359	31.196	3.493
A	post_api_ms	200	11.458	9.847	7.113	8.721	12.275	21.194	27.426	6.645	33.856	4.598
A	sandbox_create_seconds	6	0.914	0.925	0.824	0.875	0.938	1.004	1.022	0.811	1.026	0.067
B	get_trusted_ms	200	17.881	17.069	13.119	14.864	20.166	26.033	30.692	11.430	43.457	4.337
B	post_api_ms	200	22.653	20.829	16.048	18.229	25.634	34.197	40.642	14.422	65.327	6.601
B	sandbox_create_seconds	6	0.986	0.995	0.917	0.955	1.010	1.046	1.056	0.908	1.058	0.048

5. Findings about the components

F1. OpenShell L7 rules default to audit (C3b-A). An endpoint with L7 rules and no enforcement field logs a violation and forwards the request. We observed `POST /admin/delete` reaching the upstream despite a rule allowing only `POST /v1/**`. The OCSF event reads `ALLOWED ... engine:l7`. OpenShell documents this default. Operators who write rules and never set `enforcement: enforce` get no enforcement.

BULL now refuses such policies at `CreateSandbox`, `UpdateConfig` and rule merges (`bull_l7_audit_mode`). The check accepts both the YAML form (`enforce`) and the protobuf-JSON enum form (`NETWORK_ENFORCEMENT_MODE_ENFORCE`); `UNSPECIFIED` is treated as audit.

F2. Transparent interception holds (C8). OpenShell's policy DNS resolved `host.openshell.internal` to a trap address (198.18.0.2). Traffic sent with `--noproxy` was still captured, L7-evaluated and, in phase B, sent through BULL. Binary identity refused the Python client (EACCES) even though curl was allowed to the same endpoint. Neither a raw socket to a public IP nor UDP DNS got out.

F3. Middleware metadata gaps in v0.1.2.

- `HttpRequestTarget.scheme` is set to `https` on relay paths that carry plaintext HTTP. OpenShell's own OCSF events for the same requests say `http`.
- `RequestContext.originating_process` is hard-coded to `None` for operator middleware, so BULL cannot bind decisions to the executable. Every binary field in the BULL ledger is empty.

BULL therefore must not trust scheme; the correlator compares only host, port and path.

F4. Extension contract is strict and fails closed. Two mistakes each blocked the path completely rather than opening it:

- Replies whose `reason_code` does not match `[a-z][a-z0-9_]*` are treated as middleware failure (`response_reason_code_invalid`), and the request is denied.
- An extension that omits the `openshell.<family>.contract` capability is refused at gateway start.

F5. OCSF JSONL export depends on a writable `/var/log` in the supervisor container. It is not writable in the Docker driver's image, so enabling `ocsf_json_enabled` produced no file. The same events are available as shorthand through the gateway (`openshell logs`), which is what C10 uses.

F6. `openshell sandbox exec` reads a non-TTY stdin to EOF before dispatching. Automation that inherits an open stdin hangs indefinitely. Pass `/dev/null`.

6. Defects found and fixed in BULL during the experiment

All six were invisible to BULL's unit tests and surfaced only against the live gateway. Each fix has a regression test (`tests/test_openshell_authority.py`, 28 tests).

#	Defect	Symptom on the live system	Fix
1	Missing contract capabilities in <code>PeerMetadata</code>	Gateway refused to start	Declare <code>openshell.<family>.contract</code> as supported and required
2	Dotted reason codes (<code>bull.denied</code>)	Every request denied as <code>middleware_failed: response_reason_code_invalid</code> , fail-closed but useless	Codes renamed to <code>bull_*</code> ; grammar test added

#	Defect	Symptom on the live system	Fix
3	Protobuf-JSON ports are doubles (40605.0)	Endpoint keys host:40605.0 would never match later host:40605 checks, a latent false-widening or bypass	Normalize to int; also cover every entry in ports
4	No audit-mode check	OpenShell-deny cases silently became allow (F1)	bull_l7_audit_mode refusal
5	Enum form of enforcement not recognized	After fix 4, every sandbox was refused	Accept the NETWORK_ENFORCEMENT_MODE_ENFORCE enum name
6	Correlator let a denied decision claim a nearby allowed request's receipt	C10 reported a false "effect without allow"	Effects assigned only to allowed decisions at or after decision time; leftovers are the violation signal

7. Threats to validity and limitations

- **Unauthenticated extension transport.** The middleware ran over plaintext gRPC with `allow_insecure_transport`, and OpenShell warns that the service "cannot distinguish OpenShell from any other network client". Any process on the bridge could have queried BULL's verdicts, though not changed an enforcement outcome. Production needs https with gateway-JWT validation. We did not test that configuration.
- **Single host, Docker driver only.** No Kubernetes, VM or hardware-isolated drivers. Numbers come from a 4-vCPU shared cloud container and include its noise.
- **Exec is not mediated by BULL.** BULL sees HTTP egress and control-plane RPCs, not process execution inside the sandbox. OpenShell's Landlock, seccomp and binary identity contain exec (F2), but BULL does not authorize it. The "exec" half of the architecture's C7 is therefore OpenShell's guarantee, not BULL's.
- **WebSocket traffic is not mediated.** BULL does not bind that operation.
- **C6 phase A is weak evidence.** The provider attachment was still `waiting_for_supervisor` when the environment was checked, so absence of the canary in phase A does not show that OpenShell's placeholder mechanism works. Phase B's result rests on BULL refusing the attach, not on placeholders.
- **Correlation is heuristic.** OCSF shorthand carries no request id, so events are joined by sandbox, method, target and a 3 s window, and effects by method, target and time. The join was exact in this run, where requests were serialized. Under concurrency, identical requests could be swapped. That would not change the counts the property is judged on, but it would change row attribution. OCSF JSONL with request ids (F5) would remove the ambiguity.
- **Few runs.** Three complete runs were made. The first surfaced defect 5 (phase B could not start). In the last two, every case outcome was identical except C10, whose earlier failure was defect 6 in the correlator, not the system. Latency figures come from the final run only.
- **Approvals.** Approval used BULL's local operator socket. Human latency is not part of any measurement.

8. Conclusion

On a live OpenShell v0.1.2 gateway, BULL composed as a narrowing-only layer:

- Every effect needed both systems' consent.
- BULL's absence failed closed at both the data plane and the control plane.
- Provenance-based escalation stopped exfiltration after untrusted input, which OpenShell's static policy allowed in the baseline.
- Unauthorized widening and credential attachment were refused.
- The two systems' records joined into one consistent trail.

The main cost is about 8-11 ms per request in this build, much of it attributable to BULL's synchronous $O(n)$ ledger verification. The main risk we found is not in the composition but in a default: OpenShell's audit-mode L7 rules. BULL now refuses them.

Reproduction

```
# OpenShell v0.1.2 source at $OS, built:
# cargo build --release -p openshell-cli -p openshell-gateway -p openshell-supervisor
# cargo build --release --target x86_64-unknown-linux-musl -p openshell-sandbox
echo "172.17.0.1 host.openshell.internal" >> /etc/hosts
pip install -e '.[openshell]'
python tools/openshell_experiment.py --os-src "$OS" \
  --output docs/evidence/openshell-20260928 --latency-n 200
```

Artifacts in the evidence directory:

- results.json: all cases, metadata, latency summaries.
- cases.csv, latency.csv, latency-raw.json.
- correlation.json: C10 join, orphan events, unexplained effects.
- bull-audit.jsonl: BULL hash-chained ledger, phase B.
- ocsf-*-{A,B}.txt: OpenShell OCSF events per sandbox, as served by the gateway.
- upstream-receipts-{A,B}.json: what the destinations received.
- gateway-{A,B}.**{toml,log}**, bull.log, policy.yaml.

Appendix A. Complete case results (cases.csv)

Ph.	Case	Property	Expected	Observed	Pass
A	setup	create coder	created	ok	yes
A	setup	create reader	created	ok	yes
A	C1	BULL ALLOW + OpenShell ALLOW executes	reached upstream	http 200; reached=True	yes
A	C2	BULL DENY + OpenShell ALLOW is blocked	reached (no BULL)	http 200; reached=True	yes
A	C3	BULL ALLOW + OpenShell DENY is blocked	not reached	http 403; reached=False	yes
A	C3b	L7 rules in default audit mode (OpenShell only)	observe: violation allowed	created=True; admin POST reached=True	yes
A	C4	Escalation without BULL	reached (no approval concept)	http 200; reached=True	yes
A	C7	Internet-derived content -> POST without BULL	reached	reached=True	yes
A	C5	Policy widening without BULL	applied	✓ Policy version 3 submitted (hash: de96f00b071a) ✓ Policy version 3 loaded (active version: 3)	yes
A	C8	Native tool bypass (no proxy, other binary, raw socket, DNS) contained	no unmediated effect; raw socket and DNS fail	{"noproxy_direct": {"rc": 0, "out": "\"{\\stored\\\":true}\""}, "python_socket": {"rc": 1, "out": "raise exceptions[0]\\n File \\\"/usr/lib/python3.12/socket.py\\\", line 837, in create_connection\\n sock.connect(sa)\\nPermissionError: [Errno 13] Permission denied\""}, "python_http": {"rc": 1, "out": "File \\\"/usr/lib/python3.12/urllib/request.py\\\", line 1347, in do_open\\n raise URLError(err)\\nurllib.error.URLError: <urlopen error [Errno 13] Permission denied>\""}, "dns_udp": {"rc": 1, "out": "Traceback (most recent call last):\\n File \\\"<string>\\\", line 1, in <module>\\nOSError: [Errno 89] Destination address required\""}, "/v1/x": {"reached_upstream": true, "ocsf_engines": ["17"]}, "/v1/y": {"reached_upstream": false, "ocsf_engines": []}}	yes
A	C6	Provider credential not readable (OpenShell placeholder)	secret not in sandbox env	provider_created=True (✓ Created provider canary); attach rc=0 coder / canary: persisted (waiting_for_supervisor) Receipt: 35fb08f6-612a-452f-8b36-c2987b4f427d Persisted: 2026-09-28T18:12:43.381635941Z; secret_visible=False	yes
A	C10	Audit trail without BULL (OpenShell OCSF only)	observe: OCSF events, no authority decision record	{"ocsf_http_events": {"17": 8, "middleware": 0}, "upstream_receipts": 7}	yes
B	setup	create coder	created	ok	yes
B	setup	create reader	created	ok	yes
B	C1	BULL ALLOW + OpenShell ALLOW executes	reached upstream	http 200; reached=True	yes
B	C2	BULL DENY + OpenShell ALLOW is blocked	not reached	http 403; reached=False	yes
B	C3	BULL ALLOW + OpenShell DENY is blocked	not reached	http 403; reached=False	yes
B	C3b	L7 rules in default audit mode are refused at creation	create refused (bull_17_audit_mode)	created=False; Error: × code: 'The caller does not have permission to execute the specified operation', message: 'L7 rules must set enforcement: enforce: ['api:host.openshell.internal', 'docs:host.openshell.internal', 'news:host.openshell.internal']"	yes
B	C4	BULL ESCALATE: no effect before approval; exactly one after	0 before, 1 after approval, 0 on reuse	{"first_http": "403", "pending": 1, "approved": true, "retry_http": "200", "third_http": "403", "upstream_count": 1}	yes

Ph.	Case	Property	Expected	Observed	Pass
B	C7	Internet-derived content -> POST hits BULL provenance control	escalated	first attempt http 403, reached=False	yes
B	C5	Unauthorized policy widening rejected	rejected pending approval	Error: × code: 'The caller does not have permission to execute the specified operation', message: 'approval d029e1ad44e44632 required to widen policy with ['host.openshell.internal:9']"	yes
B	C5b	Sandbox with host outside BULL ceiling refused	rejected	Error: × code: 'The caller does not have permission to execute the specified operation', message: 'policy grants hosts outside the signed BULL ceiling: ['attacker.example']"	yes
B	C8	Native tool bypass (no proxy, other binary, raw socket, DNS) contained	no unmediated effect; raw socket and DNS fail	{"noproxy_direct": {"rc": 0, "out": "enshell.internal\\\\"layer\\\\"17\\\\"method\\\\"POST\\\\"middleware\\\\"bull-governance\\\\"path\\\\"v1/x\\\\"policy\\\\"api\\\\"port\\":35911\\\\"reason_code\\\\"bull_approval_required\\\\""}, "python_socket": {"rc": 1, "out": "raise exceptions[0]\n File \\\n/usr/lib/python3.12/socket.py\\", line 837, in create_connection\n sock.connect(sa)\nPermissionError: [Errno 13] Permission denied"}, "python_http": {"rc": 1, "out": "File \\\n/usr/lib/python3.12/urllib/request.py\\", line 1347, in do_open\n raise URLError(err)\nurllib.error.URLError: <urlopen error [Errno 13] Permission denied>"}, "dns_udp": {"rc": 1, "out": "Traceback (most recent call last):\n File \\\n<string>\\", line 1, in <module>\nOSError: [Errno 89] Destination address required"}, "/v1/x": {"reached_upstream": false, "ocsf_engines": ["17", "middleware"]}, "/v1/y": {"reached_upstream": false, "ocsf_engines": []}}	yes
B	C6	Provider attach governed; credential never readable by the agent	attach refused (no credential.use grant); secret not in sandbox	provider_created=True (✓ Created provider canary); attach: Error: × provider attachment failed (The caller does not have permission to execute the specified operation); secret_visible=False	yes
B	C9	BULL unavailable -> egress and sandbox creation fail closed	not reached; create refused	post http 403 reached=False; create_while_down=False; after_restart_docs_http=200	yes
B	C10	OpenShell events + BULL decisions + effects correlate	every decision matched, consistent; no unexplained effect; ledger verifies	{"bull_decisions": 9, "matched_openshell_event": 9, "consistent": 9, "effects_without_bull_allow": 0, "openshell_events_without_bull_decision": 1, "orphans_fail_closed": true, "ledger_valid": true, "ledger_records": 19, "ocsf_http_events": {"17": 11, "middleware": 10}, "upstream_receipts": 5}	yes

A.1 C8 bypass probes (full evidence)

Phase	Probe	Evidence
A	noproxy_direct	{"rc": 0, "out": "{\\"stored\\":true}"}
A	python_socket	{"rc": 1, "out": "raise exceptions[0]\n File \\\n/usr/lib/python3.12/socket.py\\", line 837, in create_connection\n sock.connect(sa)\nPermissionError: [Errno 13] Permission denied"}
A	python_http	{"rc": 1, "out": "File \\\n/usr/lib/python3.12/urllib/request.py\\", line 1347, in do_open\n raise URLError(err)\nurllib.error.URLError: <urlopen error [Errno 13] Permission denied>"}
A	dns_udp	{"rc": 1, "out": "Traceback (most recent call last):\n File \\\n<string>\\", line 1, in <module>\nOSError: [Errno 89] Destination address required"}
A	/v1/x	{"reached_upstream": true, "ocsf_engines": ["17"]}
A	/v1/y	{"reached_upstream": false, "ocsf_engines": []}
B	noproxy_direct	{"rc": 0, "out": "enshell.internal\\\\"layer\\\\"17\\\\"method\\\\"POST\\\\"middleware\\\\"bull-governance\\\\"path\\\\"v1/x\\\\"policy\\\\"api\\\\"port\\":35911\\\\"reason_code\\\\"bull_approval_required\\\\"}"}
B	python_socket	{"rc": 1, "out": "raise exceptions[0]\n File \\\n/usr/lib/python3.12/socket.py\\", line 837, in create_connection\n sock.connect(sa)\nPermissionError: [Errno 13] Permission denied"}
B	python_http	{"rc": 1, "out": "File \\\n/usr/lib/python3.12/urllib/request.py\\", line 1347, in do_open\n raise URLError(err)\nurllib.error.URLError: <urlopen error [Errno 13] Permission denied>"}
B	dns_udp	{"rc": 1, "out": "Traceback (most recent call last):\n File \\\n<string>\\", line 1, in <module>\nOSError: [Errno 89] Destination address required"}
B	/v1/x	{"reached_upstream": false, "ocsf_engines": ["17", "middleware"]}

Phase	Probe	Evidence
B	/v1/y	{"reached_upstream": false, "ocsf_engines": []}

Appendix B. Audit correlation, phase B (correlation.json)

Metric	Value
bull_decisions	9
matched_openshell_event	9
consistent	9
effects_without_bull_allow	0
openshell_events_without_bull_decision	1
orphans_fail_closed	True

#	Sandbox	Method	Resource	Provenance	BULL	Allowed	Code	Approval	OS L7	OS mw	Effect	Consistent	Ledger hash
1	coder	GET	https://host.openshell.internal:40959/guide	local_trusted	ALLOW	True	bull_allowed		ALLOWED	ALLOWED	True	True	f0 added9b52d0c1f2b
2	coder	POST	https://host.openshell.internal:35911/v1/records	local_trusted	ALLOW	True	bull_allowed		ALLOWED	ALLOWED	True	True	00c66c941d0d2e24
3	reader	POST	https://host.openshell.internal:35911/v1/records	local_trusted	DENY	False	bull_denied		ALLOWED	DENIED	False	True	bfbdacde84242d5d
4	coder	GET	https://host.openshell.internal:46731/article	local_trusted	ALLOW	True	bull_allowed		ALLOWED	ALLOWED	True	True	f50244dca304d016
5	coder	POST	https://host.openshell.internal:35911/v1/records	internet	ESCALATE	False	bull_approval_required	36b9cbbcf4819059	ALLOWED	DENIED	False	True	bb0324f5168571b1
6	coder	POST	https://host.openshell.internal:35911/v1/records	internet	ESCALATE	True	bull_approved	36b9cbbcf4819059	ALLOWED	ALLOWED	True	True	804e34a6b6c0f4a5
7	coder	POST	https://host.openshell.internal:35911/v1/records	internet	ESCALATE	False	bull_approval_required	6452be302b3d2906	ALLOWED	DENIED	False	True	c9ef970549c50fca
8	coder	POST	https://host.openshell.internal:35911/v1/x	internet	ESCALATE	False	bull_approval_required	fe51d8900f4821ec	ALLOWED	DENIED	False	True	8f9238c1744bd7d2
9	coder	GET	https://host.openshell.internal:40959/after-restart	internet	ALLOW	True	bull_allowed		ALLOWED	ALLOWED	True	True	82b54aba3d8d430a

B.1 OpenShell events with no BULL decision

Time	Sandbox	Method	URL	Action	Reason
1790619185.513	coder	POST	http://host.openshell.internal:35911/v1/records	DENIED	middleware_failed: external_service_error

Appendix C1. OpenShell OCSF HTTP events, phase A (ocsf-*-A.txt)

Epoch (s)	Sandbox	Engine	Action	Method	URL	Policy	Reason
1790619153.612	code r	l7	ALLOWED	GET	http://host.openshell.internal:40959/guide	docs	
1790619153.667	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619153.734	reader	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619153.793	code r	l7	DENIED	POST	http://host.openshell.internal:35911/admin/delete	api	L7_REQUEST deny POST host.openshell.internal:35911/admin/delete reason=POST /admin/delete not permitted by policy
1790619154.773	auditor	l7	ALLOWED	POST	http://host.openshell.internal:35911/admin/delete	api	
1790619154.822	code r	l7	ALLOWED	GET	http://host.openshell.internal:46731/article	news	
1790619154.883	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619162.980	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/x	api	

Upstream receipts, phase A (upstream-receipts-A.json)

Epoch (s)	Method	URL
1790619153.613	GET	http://host.openshell.internal:40959/guide
1790619153.668	POST	http://host.openshell.internal:35911/v1/records
1790619153.735	POST	http://host.openshell.internal:35911/v1/records
1790619154.775	POST	http://host.openshell.internal:35911/admin/delete
1790619154.827	GET	http://host.openshell.internal:46731/article
1790619154.884	POST	http://host.openshell.internal:35911/v1/records
1790619162.980	POST	http://host.openshell.internal:35911/v1/x

Appendix C2. OpenShell OCSF HTTP events, phase B (ocsf-*-B.txt)

Epoch (s)	San dbox	Engine	Action	Method	URL	Policy	Reason
1790619184.386	code r	l7	ALLOWED	GET	http://host.openshell.internal:40959/guide	docs	
1790619184.390	code r	middleware	ALLOWED	GET	http://host.openshell.internal:40959/guide	docs	
1790619184.437	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619184.442	code r	middleware	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619184.504	reader	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619184.509	reader	middleware	DENIED	POST	http://host.openshell.internal:35911/v1/records	api	middleware_denied:bull-governance:bull_denied
1790619184.567	code r	l7	DENIED	POST	http://host.openshell.internal:35911/admin/delete	api	L7_REQUEST deny POST host.openshell.internal:35911/admin/delete reason=POST /admin/delete not permitted by policy
1790619184.645	code r	l7	ALLOWED	GET	http://host.openshell.internal:46731/article	news	
1790619184.649	code r	middleware	ALLOWED	GET	http://host.openshell.internal:46731/article	news	
1790619184.704	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619184.709	code r	middleware	DENIED	POST	http://host.openshell.internal:35911/v1/records	api	middleware_denied:bull-governance:bull_approval_required
1790619184.785	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619184.790	code r	middleware	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619184.845	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619184.852	code r	middleware	DENIED	POST	http://host.openshell.internal:35911/v1/records	api	middleware_denied:bull-governance:bull_approval_required
1790619184.957	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/x	api	
1790619184.962	code r	middleware	DENIED	POST	http://host.openshell.internal:35911/v1/x	api	middleware_denied:bull-governance:bull_approval_required
1790619185.512	code r	l7	ALLOWED	POST	http://host.openshell.internal:35911/v1/records	api	
1790619185.513	code r	middleware	DENIED	POST	http://host.openshell.internal:35911/v1/records	api	middleware_failed: external_service_error
1790619185.877	code r	l7	ALLOWED	GET	http://host.openshell.internal:40959/after-restart	docs	
1790619185.884	code r	middleware	ALLOWED	GET	http://host.openshell.internal:40959/after-restart	docs	

Upstream receipts, phase B (upstream-receipts-B.json)

Epoch (s)	Method	URL
1790619184.391	GET	http://host.openshell.internal:40959/guide
1790619184.443	POST	http://host.openshell.internal:35911/v1/records
1790619184.650	GET	http://host.openshell.internal:46731/article
1790619184.791	POST	http://host.openshell.internal:35911/v1/records
1790619185.885	GET	http://host.openshell.internal:40959/after-restart

Appendix D. Raw latency dataset (latency-raw.json)

This is every sample, in run order. Request latencies are `curl time_total` in ms, measured inside the bench sandbox.

#	A GET	B GET	A POST	B POST
1	18.761	28.393	8.973	22.982
2	12.082	13.398	9.828	17.536
3	9.509	16.153	10.169	16.234
4	9.405	16.351	8.196	16.635
5	8.891	21.402	9.279	20.731
6	8.779	18.261	15.123	16.020
7	11.115	13.653	17.178	20.223
8	8.049	13.316	27.422	16.974
9	13.269	15.529	20.176	19.716
10	9.684	13.927	10.448	30.055
11	10.325	17.886	10.835	21.414
12	7.824	22.099	10.167	27.099
13	16.403	25.889	10.569	19.259
14	12.752	13.362	11.420	15.195
15	23.452	14.859	9.379	19.972
16	8.921	15.283	9.203	15.515
17	10.346	13.993	10.329	17.309
18	7.776	13.853	10.335	16.000
19	8.739	12.642	10.983	21.327
20	13.088	13.800	27.796	18.089
21	7.811	16.106	9.571	24.082
22	10.529	16.198	14.016	19.712
23	8.137	13.453	12.275	15.761
24	9.140	20.253	11.287	17.626
25	12.208	19.172	7.755	16.371
26	8.378	19.880	8.067	16.049
27	8.335	19.921	7.070	18.806
28	8.393	15.612	8.885	16.676
29	10.935	15.972	10.640	18.229
30	31.196	19.588	14.116	18.603
31	8.365	14.231	11.323	15.776
32	13.624	13.652	12.788	19.900
33	8.235	13.891	8.191	29.976
34	7.552	13.366	13.844	25.406
35	7.609	14.925	11.756	20.492
36	7.126	12.330	20.759	21.516
37	9.323	11.430	25.899	16.308
38	8.066	17.769	11.603	14.422
39	10.471	20.163	11.027	18.334
40	9.009	17.898	7.586	20.783
41	8.558	14.865	6.908	22.535
42	10.346	14.259	14.448	21.211
43	8.698	18.774	7.712	28.948
44	15.492	15.142	7.259	30.882
45	7.458	13.890	8.557	26.767
46	16.496	13.207	7.157	23.907
47	16.190	12.685	8.424	20.139

#	A GET	B GET	A POST	B POST
101	8.071	17.330	9.711	17.383
102	6.943	43.457	8.507	20.299
103	7.253	27.507	8.778	20.679
104	7.924	17.589	9.818	16.515
105	7.589	19.970	8.505	24.811
106	8.349	15.240	8.915	25.735
107	9.731	15.225	8.958	34.712
108	11.791	20.967	11.244	27.810
109	7.049	15.038	11.374	23.515
110	6.681	16.364	10.160	17.915
111	7.226	17.535	13.773	20.919
112	6.468	24.888	13.641	19.563
113	7.792	26.118	25.978	21.430
114	6.842	16.642	12.016	18.614
115	8.497	22.623	9.829	22.709
116	12.504	18.336	7.558	36.769
117	10.670	18.693	10.093	17.720
118	14.685	14.288	9.650	19.729
119	7.340	15.888	9.443	22.476
120	11.336	14.875	9.346	20.542
121	9.917	17.019	12.103	18.229
122	14.253	22.247	9.542	20.192
123	11.214	20.949	10.281	21.025
124	9.616	20.336	8.150	25.451
125	10.614	20.399	8.504	29.375
126	8.090	22.760	13.026	37.554
127	9.512	16.609	10.924	20.262
128	8.325	20.417	18.845	19.743
129	8.863	17.001	15.106	18.652
130	7.458	13.753	10.135	19.755
131	9.051	19.030	7.042	22.145
132	17.005	18.113	7.740	16.958
133	14.545	25.585	7.114	25.112
134	13.103	26.597	7.026	18.379
135	8.218	20.523	6.645	29.107
136	7.807	20.520	7.098	65.327
137	8.961	20.623	9.421	32.676
138	11.121	15.479	12.054	26.971
139	9.871	17.658	9.773	25.423
140	8.438	15.289	10.440	23.896
141	11.215	16.656	8.841	17.635
142	8.091	21.610	9.020	19.197
143	13.692	20.467	8.675	25.504
144	8.121	20.330	10.846	20.549
145	8.693	24.268	21.169	26.329
146	8.494	30.677	22.363	28.185
147	9.828	26.029	10.469	28.169

#	A GET	B GET	A POST	B POST
48	7.881	13.455	9.462	15.118
49	8.497	13.960	9.843	27.652
50	7.175	13.654	8.224	18.619
51	6.823	21.881	12.149	20.690
52	6.583	22.633	16.906	35.993
53	6.359	15.158	15.633	28.010
54	6.966	15.884	13.855	22.119
55	6.897	15.306	10.556	21.346
56	6.732	13.706	7.740	14.830
57	8.383	12.483	9.161	16.623
58	10.230	12.759	7.053	17.064
59	8.336	11.845	8.959	17.768
60	8.382	12.293	8.005	17.006
61	8.036	13.138	7.365	14.476
62	7.495	12.373	9.501	18.765
63	10.908	18.720	15.446	21.850
64	11.118	19.909	24.221	40.497
65	9.298	24.941	13.106	18.139
66	9.606	19.579	12.276	19.980
67	10.025	19.171	17.220	16.153
68	7.708	14.775	12.482	16.229
69	7.405	13.444	19.582	16.137
70	7.681	14.092	17.862	18.762
71	8.181	14.331	8.999	16.922
72	6.396	14.197	10.379	16.094
73	8.128	14.365	9.689	21.216
74	8.061	13.667	10.118	22.156
75	7.120	13.519	9.125	27.987
76	11.115	17.350	9.606	28.316
77	7.356	22.122	8.946	18.191
78	7.745	20.543	9.429	23.746
79	7.944	19.535	7.856	20.253
80	13.767	17.324	16.861	20.557
81	7.838	16.620	11.333	21.830
82	16.916	15.242	14.477	19.825
83	15.090	15.592	10.820	17.941
84	10.542	17.801	22.191	27.223
85	8.770	17.586	12.048	20.390
86	10.895	14.006	11.372	21.557
87	8.112	12.416	9.138	20.557
88	7.309	15.695	8.885	16.130
89	8.175	18.425	17.580	17.707
90	10.809	17.302	9.822	17.040
91	8.637	21.284	9.967	16.369
92	14.036	18.601	8.776	17.771
93	7.269	15.815	9.631	18.958
94	8.542	17.898	13.989	18.165
95	8.114	15.616	18.521	17.790
96	9.433	15.634	14.727	21.731
97	10.372	14.632	10.972	39.468
98	19.423	17.883	8.511	29.163
99	14.908	13.316	33.856	22.711
100	21.481	15.140	9.845	18.540

#	A GET	B GET	A POST	B POST
148	12.116	19.061	9.241	20.882
149	20.079	19.759	7.436	21.861
150	11.130	17.533	10.500	17.524
151	8.513	16.252	7.457	18.400
152	9.151	16.823	11.326	22.890
153	8.381	15.994	9.706	26.394
154	8.388	18.002	13.209	28.886
155	11.850	17.657	11.032	28.153
156	11.451	17.459	7.287	29.185
157	7.988	26.985	9.482	26.337
158	7.639	20.177	7.413	19.426
159	7.564	19.360	7.582	18.346
160	8.471	15.288	19.453	21.920
161	8.723	17.825	11.950	20.278
162	8.909	16.225	18.730	24.008
163	10.005	14.169	8.570	19.676
164	9.490	14.315	9.196	32.797
165	11.326	14.366	8.901	26.548
166	12.871	22.275	8.736	24.332
167	10.502	21.282	9.533	23.036
168	8.095	23.506	9.894	25.064
169	7.385	20.225	7.770	26.089
170	7.116	19.980	7.092	19.509
171	8.575	16.198	10.458	18.261
172	9.085	17.118	7.042	34.170
173	13.911	17.859	7.297	54.996
174	8.132	14.874	7.630	20.438
175	13.108	15.021	7.679	21.329
176	8.413	16.825	6.765	21.003
177	9.282	15.678	13.433	25.784
178	7.923	23.702	13.665	25.384
179	13.546	32.220	20.082	17.963
180	11.853	21.489	15.195	25.601
181	17.836	20.372	7.894	37.606
182	16.940	16.471	7.681	29.374
183	7.790	16.152	7.764	33.889
184	10.549	14.339	9.848	23.167
185	9.878	15.723	9.077	26.266
186	15.538	14.138	17.198	28.500
187	12.587	19.041	7.772	20.875
188	9.299	18.724	8.839	28.133
189	7.095	22.936	8.567	31.442
190	12.226	25.781	8.766	32.215
191	14.394	20.135	10.899	21.311
192	9.380	20.444	7.662	23.127
193	12.771	16.938	25.801	26.049
194	23.381	21.944	19.047	20.329
195	12.467	16.960	21.662	25.188
196	11.961	17.676	9.211	26.643
197	12.553	17.142	10.449	24.594
198	8.573	17.640	11.746	39.690
199	12.377	27.050	11.336	18.292
200	10.308	30.317	9.099	18.214

D.1 Sandbox creation samples (s)

#	A	B
1	1.026	1.01
2	0.935	0.908
3	0.939	1.009
4	0.914	0.946
5	0.862	1.058
6	0.811	0.982

D.2 L7 → middleware gap samples (ms)

4, 5, 4, 5, 5, 7, 5, 1, 7, 5

Appendix E. Evidence file integrity (SHA-256)

These hashes are of the files in docs/evidence/openshell-20260928/ at build time. Every number in this PDF was computed from these exact files.

File	Bytes	SHA-256
README.md	1396	d4e8adae9887f56edb225a7d8114171c71bbc66671dfc924fe7f3bad65390218
bull-audit.jsonl	312291	340a7cc304af900367c1ea221979cb81eab7f7808d4009f0ac067560669bd242
bull.log	52745	8a1cbc68fd43d3cb92a416856a0f6b5fe1e9b4347402bb32ef383b0ed3bdeda7
cases.csv	6404	23c4de9120031d4943246a07991e2f76b80a23c654049508d7df6d69101ff3d3
correlation.json	6496	c718ac4013c8a417fad372a432c0ee43c2df65e53989910653c1c1a608a4d666
gateway-A.log	242308	9e8d586e4db5ea8a3347e92b487c0454162f2e92f76261c0a214a925b7659a0b
gateway-A.toml	496	6ffdc55cfc57e629cd0a6e64f36115b2a15866d949e34ad9d94f6788b6d88f3c
gateway-B.log	261142	93b8de8ab71267cd32867087d7ee2430d1e5957a09ad3a78587224ce4b02621d
gateway-B.toml	1628	fcc36360b7265a4c24b77f5def645cb51e3822774da0f68e3ef020ad96d79398
latency-raw.json	6662	f3c7d17932a1c567d9ad2003c1b356cf0917cfc0a8ddcd045746094fdd77f56c
latency.csv	503	8c20b41183132722be2dbdb08bd0e840bb31392b616767ae200653d5ea04dac5
ocsf-auditor-A.txt	3463	4aa278f9af100986efa226cc29f019e7b8ed119f2d27758ee4b8010e0e371d22
ocsf-auditor-B.txt	0	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
ocsf-coder-A.txt	11799	69ac92422596e1c0123b11929a4bd077d405695cd2e3aff847b5185820703d43
ocsf-coder-B.txt	15885	990e98f0521ebfd1a4d30b4ff2791644d9e20d4bf6cd53461061da92ff194e1f
ocsf-reader-A.txt	3458	48e6de30cece45547ca46bddd46f072db52d3ac4ab2bc3713c4f9f847c7a6e6
ocsf-reader-B.txt	3148	8f6c8810952f285096b751e1c44edd2009f1653663805a81ab9c9835c362d35b
policy.yaml	977	c77976ada5d58555020934678b4f8e05f2061301ef82851c5130e3bdaf7053c7
results.json	17122	51e0f0c49c85cb296f93af6abdd4836b15c2bb54ce37d4485e24964531c81358
upstream-receipts-A.json	798	51db4130544dfb67d84530bbfb4f177bc89e8451f1f18cf316d9f1e88961648f
upstream-receipts-B.json	574	fdbe5c9ef8862eda1fae1093302b6aafabbe135334a5170eb853ff4f431158df

E.1 Environment (results.json)

Item	Value
OpenShell	{"source": "github.com/NVIDIA/OpenShell", "tag": "v0.1.2", "cli_version": "openshell 0.1.2"}
Host	{"kernel": "6.18.44-fc-v37", "docker": "Docker version 29.3.1, build c2be9cc"}
Result format	bull-openshell-experiment-v1

Built from repository commit a3cba78.